

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-EA-5601-04 Sheet 1 of 1
OPL Title	Thermal-Shock Avoidance on Start-Up	Discipline	Process / Operations	
Equipment	EA-5601 - SOLVENT HEATER	Area / Unit	5600 - SOLVENT HEATING SYSTEM	
Related Interlock	N/A (control loop only) (N/A)	P&ID Ref	TJC-LLD-PID-5601	
Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **thermal-shock avoidance on start-up** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Admitting steam onto a cold tube bundle too fast cracks the tube-to-tubesheet joints, causing internal leaks. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O₂ / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Establish solvent (tube-side) flow first	Permit valid & briefed
2	Admit steam slowly by ramping the TV-5602 / TIC-5602 set-point	Parameter within limit
3	Warm the exchanger through gradually	Interlock status confirmed
4	Avoid sudden large steps in the outlet temperature set-point	Change logged on DCS
5	Watch for knocking that indicates condensate hammer	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- Admitting steam onto a cold tube bundle too fast cracks the tube-to-tubesheet joints, causing internal leaks.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.